PART 7 OF 11 · LAYER 2: FILTER THE AIR

HVAC and air filtration

THE SIMPLE VERSION

Filters are like nets for air. They catch dust.
A good net has to fit tight, or the dust sneaks around it.
A small air cleaner in the bedroom helps while you sleep.

- **Filter rating.** Use at least MERV 13, or as high as your system fan and filter slot can handle. Have your HVAC technician confirm what your equipment can accept.[1]
- **Filter fit.** A high-rated filter does little if air bypasses it. It should sit snug in its slot or grille with no visible gaps. Overloaded filters also lose airflow and increase bypass, so change them on schedule.[2]
- **Know when the filter works.** A central filter only cleans air while the system runs, which is usually less than 25 percent of the time.[1] We do not run the fan continuously in Florida because of the humidity penalty, so a bedroom HEPA unit fills the gap.
- **A filter that is fully coated every month is a clue.** It often means leaky return ductwork or a return pulling dusty attic or wall-cavity air. Ask your HVAC contractor to test duct leakage and seal returns instead of only changing filters more often.

Portable HEPA air cleaner

True HEPA is a defined standard: the filter can theoretically remove at least 99.97 percent of airborne particles 0.3 microns in size.[3] Size the unit by its clean air delivery rate (CADR), not by the box photo. EPA's estimate for 8-foot ceilings:[1]

Room size (sq ft)	100	200	300	400	500	600
Minimum CADR (cfm)	65	130	195	260	325	390

- Run it continuously while the room is occupied, place it near the bed where it can draw from the breathing zone, keep it out of corners and from behind furniture, and replace filters on schedule.[4]
- **Know its limits.** Many indoor allergens ride on large particles that settle onto surfaces quickly, so air cleaners cannot remove them well. Effective allergen control still depends on routine cleaning and dust control.[2]
- **Avoid ozone generators and ionizers.** At levels that meet public-health standards, ozone does not effectively remove mold or other biological pollutants, and no federal agency has approved ozone generators for occupied spaces.[5] Ionizer cleaners are not recommended for mite allergy because they can produce ozone, which can trigger asthma.[4]

Your next step

Check your filter size and MERV rating, and confirm it fits with no gaps. Measure the sensitive person's bedroom so you can match an air cleaner's CADR to the room.

Sources for this part

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